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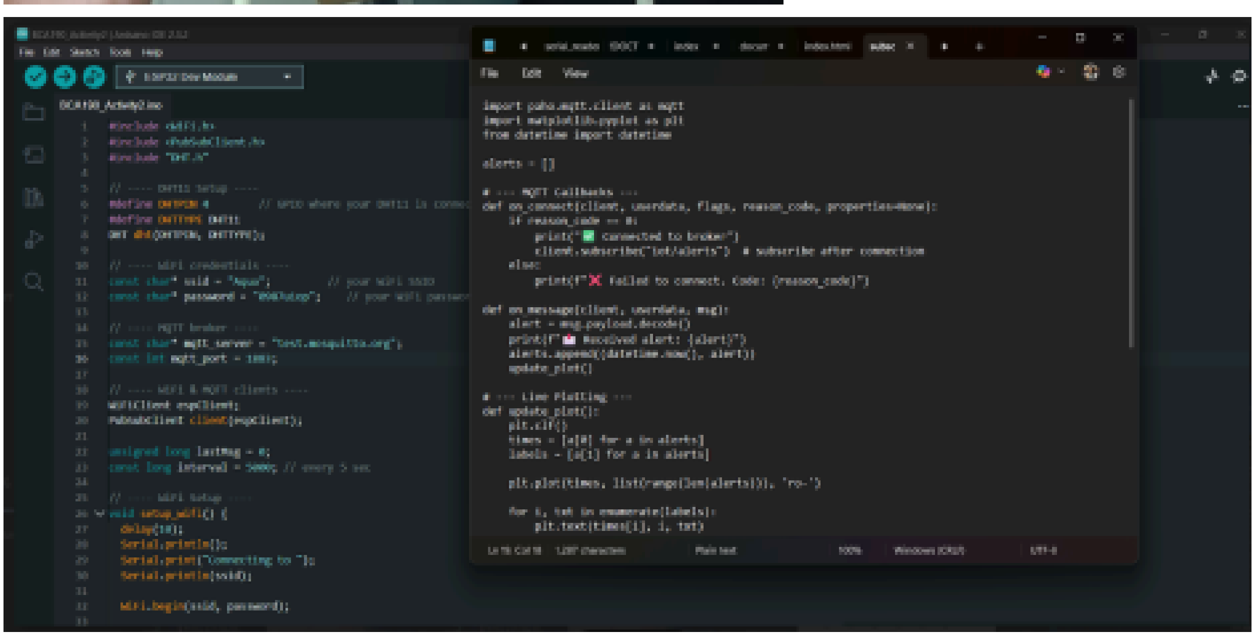
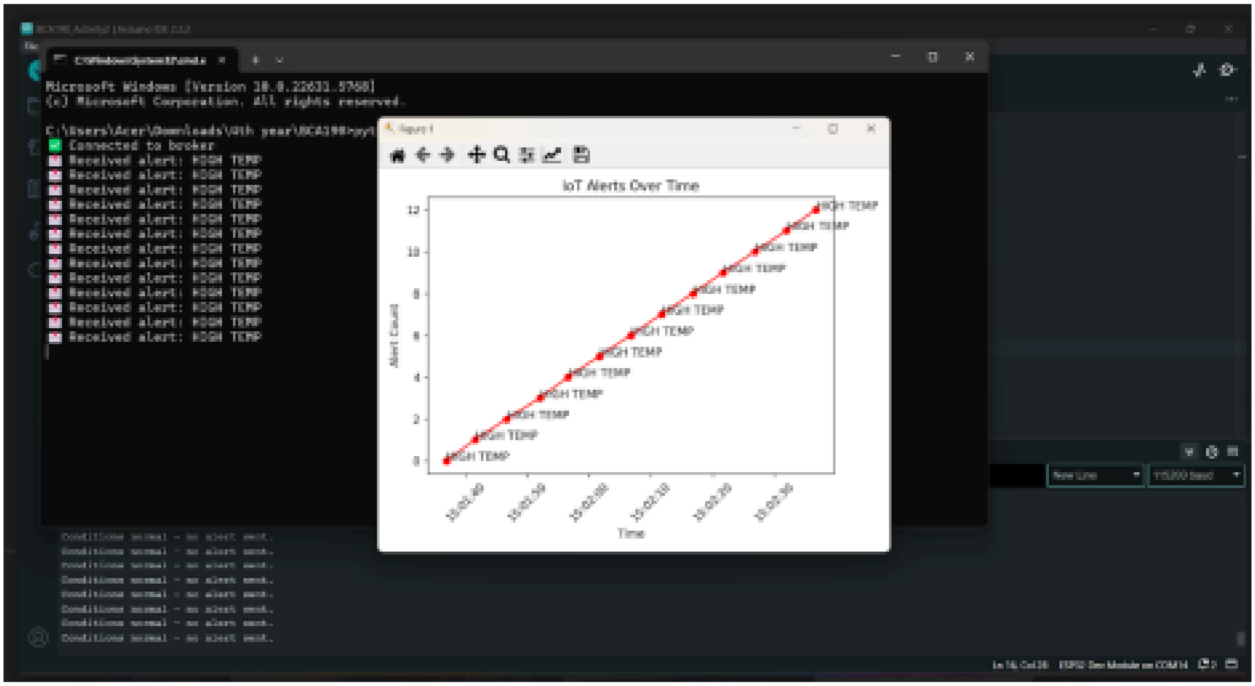
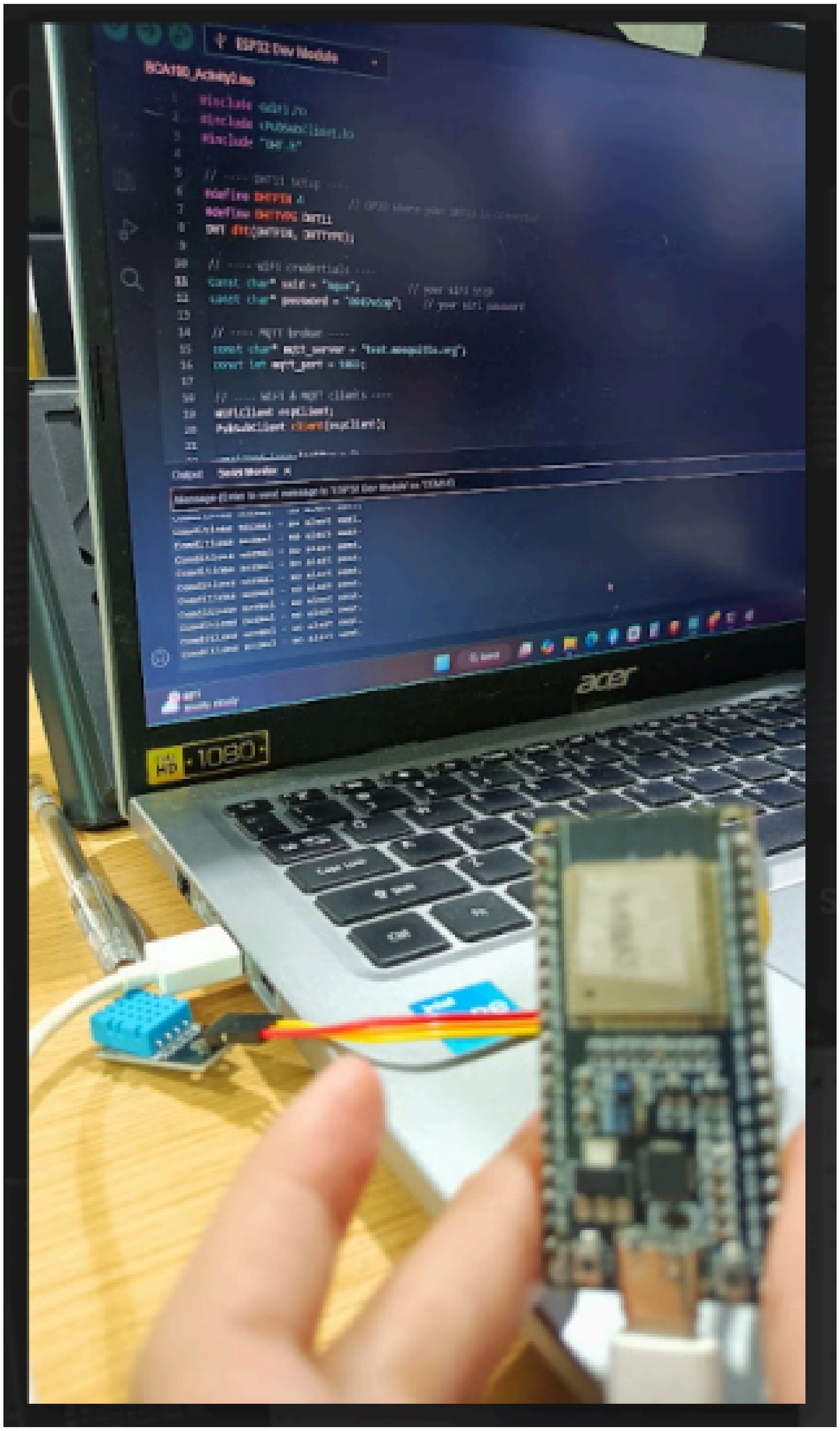
BCA190 B188.1

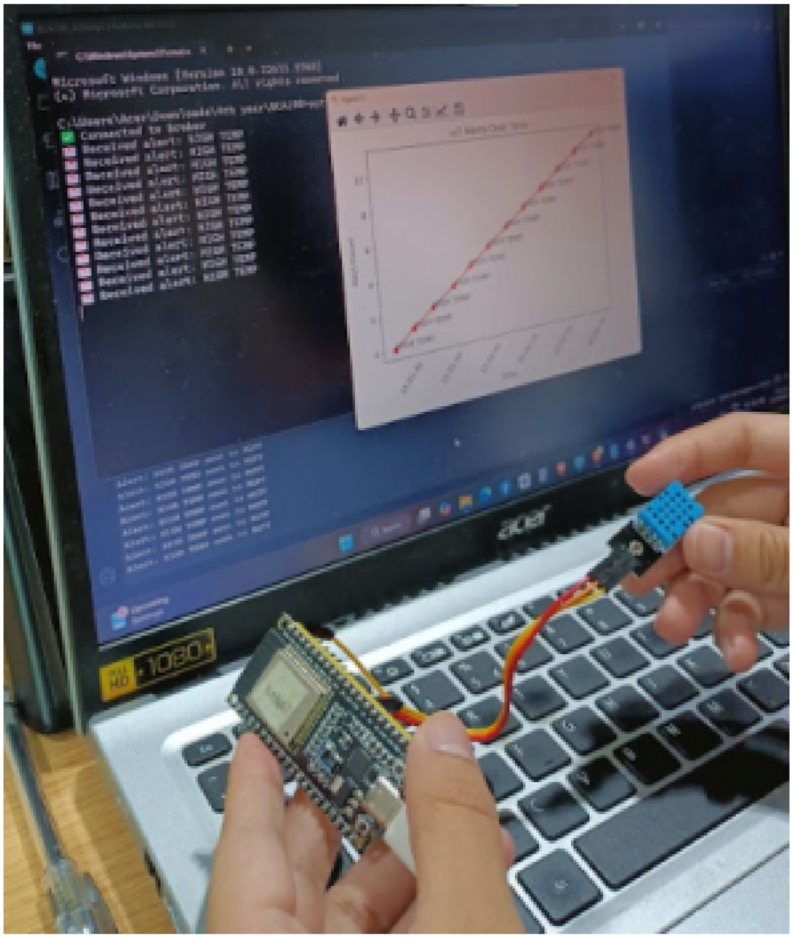
Lab Activity No. 2
Implementing Edge Computing for IoT with ESP32

Circuit Connections:




Microsoft Windows [Version 10.0.22631.5768]
(c) Microsoft Corporation. All rights reserved.
Connected to broker
Received alert: HIGH TEMP
IoT Alerts Over Time
Alert Count
Time
HIGH TEMP
Conditions normal - no alert sent.


ESP32 Dev Module
acer
HD
1080



Assessment Questions

1. What is the main advantage of processing data at the edge (ESP32) instead of sending all raw data to the cloud?

   <u>Answer:</u> The main advantage of processing data at the edge, on a device like the ESP32, is **reduced latency**. By making decisions locally, the system can react almost instantly to critical events, such as a high temperature alert, without the delay of sending data to the cloud, waiting for it to be processed, and then receiving a response. This is crucial for time-sensitive applications.
2. How does edge computing reduce bandwidth usage?

<u>Answer:</u> Edge computing reduces bandwidth usage by **minimizing data transmission**. Instead of continuously sending all raw sensor readings (like temperature and humidity data every few seconds), the ESP32 only sends a small alert message when a specific condition is met. This significantly cuts down on the amount of data transmitted, which is especially important for IoT systems that rely on cellular or other costly data plans.

3. Modify the ESP32 program to also alert when humidity is above 80%. What line of code would you change?

<u>Answer:</u> To modify the ESP32 program to also alert when humidity is above 80%, you would add an else if condition to the existing decision logic. The specific lines of code to change are within the loop() function.

```C++
// Edge decision logic
if (t > 30) {
 client.publish("iot/alerts", "HIGH TEMP");
 Serial.println("Alert: HIGH TEMP sent to MQTT");
}
else if (h < 40) {
 client.publish("iot/alerts", "LOW HUMIDITY");
 Serial.println("Alert: LOW HUMIDITY sent to MQTT");
}
else if (h > 80) { // New condition
 client.publish("iot/alerts", "HIGH HUMIDITY");
 Serial.println("Alert: HIGH HUMIDITY sent to MQTT");
}
else {
 Serial.println("Conditions normal - no alert sent.");
}
```

4. In which real-world IoT applications would edge computing be critical (e.g., healthcare, autonomous vehicles)?

   Answer: Edge computing is critical in real-world IoT applications like autonomous vehicles and healthcare because it enables low-latency decision-making. In these scenarios, a delay of even a fraction of a second can have severe consequences, making it essential to process data and react instantly on the device itself rather than relying on a distant cloud server.

5. Compare the trade-offs between cloud-only IoT vs. edge + cloud hybrid IoT.

   Answer: The main trade-off between cloud-only IoT and an edge + cloud hybrid model is the balance between latency/reliability and computational power/cost. Cloud-only systems offer immense processing and storage capabilities but suffer from high latency and are vulnerable to internet outages. In contrast, a hybrid approach leverages edge devices for real-time, resilient local processing, while using the cloud for long-term data analysis, system-wide management, and less time-critical tasks, leading to more efficient bandwidth use and a lower overall operating cost.